Equivalent expressions — Part 2

Advanced Math · Math · 58 questions

2.1 c0558845-d200-4e15-8b86-f37c4c3c5484 · Medium

Which of the following expressions is equivalent to $8x^{10} - 8x^9 + 88x$?

A. $x7x^{10} - 7x^9 + 87x$

B. $x8^{10} - 8^9 + 88$

C. $8xx^{10} - x^9 + 11x$

D. $8xx^9 - x^8 + 11$

2.2 2aeea10a-c9ed-426c-912a-939dedc77aac · Medium

$$fx = 2x + 3$$

$$gx = 7x - 2$$

$$hx = 5x + 6$$

The functions f, g, and h are defined as shown. If $fx \cdot gx - hx = ax^2 + bx + c$, where a, b, and c are constants, what is the value of b?

A. -5

B. 12

C. 20

D. 22

2.3 891c7c48-227c-4992-acdb-e5e19b360217 · Medium

Which expression is equivalent to $7x^3 + 7x - 6x^3 - 3x$?

A. $x^3 + 10x$

B. $-13x^3 + 10x$

C. $-13x^3 + 4x$

D. $x^3 + 4x$

2.4 022343-DC · Medium

$(2x+3)-(x-7)$

Which of the following is equivalent to the given expression?

A. $x-4$

B. $3x-4$

C. $x+10$

D. $2x^2+21$

2.5 029490-DC · Medium

$(4x^3-5x^2+3)-(6x^3+2x^2-x)$

Which of the following expressions is equivalent to the expression above?

A. $-10x^3-3x^2+x+3$

B. $-2x^3-7x^2+x+3$

C. $-2x^3-3x^2+x+3$

D. $10x^3-7x^2-x+3$

2.6 74b2dded-7f3f-4e1c-ba2e-dada922f008c · Medium

Which expression is equivalent to $8x^3 + 8 - x^3 - 2$?

A. $8x^3 + 6$

B. $7x^3 + 10$

C. $8x^3 + 10$

D. $7x^3 + 6$

2.7 d228702c-741e-49d7-ad2d-527046f58b94 · Medium

Which expression is equivalent to $\frac{8xx - 7 - 3x - 7}{2x - 14}$, where $x > 7$?

A. $\frac{x - 7}{5}$

B. $\frac{8x - 3}{2}$

C. $\frac{8x^2 - 3x - 14}{2x - 14}$

D. $\frac{8x^2 - 3x - 77}{2x - 14}$

2.8 00664-DC · Medium

$x^2 + 6x + 4$

Which of the following is equivalent to the expression above?

A. $(x + 3)^2 + 5$

B. $(x + 3)^2 - 5$

C. $(x - 3)^2 + 5$

D. $(x - 3)^2 - 5$

2.9 08064-DC · Medium

Which of the following expressions is equivalent to x^2-5 ?

A. $(x+\sqrt{5})^2$

B. $(x-\sqrt{5})^2$

C. $(x+\sqrt{5})(x-\sqrt{5})$

D. $(x+5)(x-1)$

2.10 07749-DC · Medium

If $x^2=a+b$ and $y^2=a+c$, which of the following is equal to $(x^2-y^2)^2$?

A. $a^2-2ac+c^2$

B. $b^2-2bc+c^2$

C. $4a^2-4abc+c^2$

D. $4a^2-2abc+b^2c^2$

2.11 070623-DC · Medium

$$\left(\frac{1}{2}x+\frac{3}{2}\right)\left(\frac{3}{2}x+\frac{1}{2}\right)$$

The expression above is equivalent to ax^2+bx+c, where a, b, and c are constants. What is the value of b?

Student-produced response: _____

2.12 08126-DC · Medium

$\sqrt[3]{x^3y^6}$

Which of the following expressions is equivalent to the expression above?

A. y^2

B. xy^2

C. y^3

D. xy^3

2.13 94498a74-a763-40c5-b248-608f7ff7f6e4 · Medium

Which expression is equivalent to $\frac{4x+1}{x-9}$, where $x > 9$?

A. $4 + \frac{37}{x-9}$

B. $4 + \frac{35}{x-9}$

C. $4 + \frac{1}{x-9}$

D. $4 - \frac{1}{x-9}$

2.14 016475-DC · Medium

Which of the following is equivalent to the expression $x^4 - x^2 - 6$?

A. $(x^2+1)(x^2-6)$

B. $(x^2+2)(x^2-3)$

C. $(x^2+3)(x^2-2)$

D. $(x^2+6)(x^2-1)$

2.15 74a8d55a-4db1-41ce-9d72-b5498e6a1d89 · Medium

The expression $90y^5 - 54y^4$ is equivalent to $ry^4(15y - 9)$, where r is a constant. What is the value of r?

Student-produced response: _____

2.16 022234-DC · Medium

If $p = 3x + 4$ and $v = x + 5$, which of the following is equivalent to $pv - 2p + v$?

A. $3x^2 + 12x + 7$

B. $3x^2 + 14x + 17$

C. $3x^2 + 19x + 20$

D. $3x^2 + 26x + 33$

2.17 05487-DC · Medium

$$\left(\frac{1}{2}x + 3\right) - \left(\frac{2}{3}x - 5\right)$$

Which of the following is equivalent to the expression above?

A. $-\frac{1}{6}x + 8$

B. $-\frac{1}{6}x - 2$

C. $-\frac{1}{3}x^2 + \frac{1}{2}x + 15$

D. $-\frac{1}{3}x^2 - \frac{9}{2}x - 15$

2.18 029468-DC · Medium

Which of the following is equivalent to the sum of $3x^4+2x^3$ and $4x^4+7x^3$?

A. $16x^{14}$

B. $7x^8+9x^6$

C. $12x^4+14x^3$

D. $7x^4+9x^3$

2.19 05799-DC · Medium

$(x-11y)(2x-3y)-12y(-2x+3y)$

Which of the following is equivalent to the expression above?

A. $x-23y$

B. $2x^2-xy-3y^2$

C. $2x^2+24xy+36y^2$

D. $2x^2-49xy+69y^2$

2.20 05410-DC · Medium

Which of the following is an equivalent form of $(1.5x-2.4)^2-(5.2x^2-6.4)$?

A. $-2.2x^2+1.6$

B. $-2.2x^2+11.2$

C. $-2.95x^2-7.2x+12.16$

D. $-2.95x^2-7.2x+0.64$

2.21 09342-DC · Medium

$4a^2+20ab+25b^2$

Which of the following is a factor of the polynomial above?

A. $a+b$

B. $2a+5b$

C. $4a+5b$

D. $4a+25b$

2.22 09133-DC · Medium

The sum of $-2x^2+x+31$ and $3x^2+7x-8$ can be written in the form ax^2+bx+c, where a, b, and c are constants. What is the value of $a+b+c$?

Student-produced response: _____

2.23 3c39fa3c-f27f-4029-8473-2bb4b3ca5672 · Medium

Which expression is equivalent to $2x - \frac{9}{2}x + 8$?

A. $2x^2 - 72$

B. $2x^2 + 7x - 72$

C. $2x^2 - x - 36$

D. $x^2 + 7x - 36$

2.24 82103492-4feb-429c-a10f-a71972d8a35d · Medium

$$g(x) = \frac{3}{5}x + \frac{7}{6}$$

$$h(x) = 6x - 5$$

The functions g and h are defined by the equations shown. Which expression is equivalent to $g(x) \cdot h(x)$?

A. $\frac{18x^2}{5} - \frac{35}{6}$

B. $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$

C. $\frac{18x^2}{5} - 4x - \frac{35}{6}$

D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$

2.25 00538-DC · Hard

The expression $\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$, where $x > 1$ and $y > 1$, is equivalent to which of the following?

A. $\frac{\sqrt{y}}{\sqrt[3]{x^2}}$

B. $\frac{y\sqrt{y}}{\sqrt[3]{x^2}}$

C. $\frac{y\sqrt{y}}{x\sqrt{x}}$

D. $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

2.26 025860-DC · Hard

The equation $\frac{x^2+6x-7}{x+7}=ax+d$ is true for all $x \neq -7$, where a and d are integers. What is the value of $a+d$?

A. -6

B. -1

C. 0

D. 1

2.27 09413-DC · Hard

$f(x)=x^3-9x$

$g(x)=x^2-2x-3$

Which of the following expressions is equivalent to $\frac{f(x)}{g(x)}$, for $x>3$?

A. $\frac{1}{x+1}$

B. $\frac{x+3}{x+1}$

C. $\frac{x(x-3)}{x+1}$

D. $\frac{x(x+3)}{x+1}$

2.28 1775076a-9ccb-4b4b-982d-4765d16527dc · Hard

The expression $\frac{29x+102}{xx+51}$ is equivalent to $\frac{p}{x}+\frac{w}{x+51}$, where p and w are constants. What is the value of w?

Student-produced response: _____

2.29 070622-DC · Hard

If $\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$ for all positive values of x, what is the value of $\frac{a}{b}$?

Student-produced response: _____

2.30 44b4a3f1-1823-4c9d-984b-a4ea335c95ad · Hard

One of the factors of $2x^3 + 42x^2 + 208x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b?

Student-produced response: _____

2.31 05695-DC · Hard

Which of the following expressions is equivalent to $\frac{x^2-2x-5}{x-3}$?

A. $x - 5 - \frac{20}{x-3}$

B. $x - 5 - \frac{10}{x-3}$

C. $x + 1 - \frac{8}{x-3}$

D. $x + 1 - \frac{2}{x-3}$

2.32 aed0b60c-1ac2-47c0-8e11-047b2c3d1b5e · Hard

$$\sqrt[5]{70n}\,\sqrt[6]{70n^2}$$

For what value of x is the given expression equivalent to $70n^{30x}$, where $n > 1$?

Student-produced response: _____

2.33 11189-DC · Hard

$(ax+3)(5x^2-bx+4)=20x^3-9x^2-2x+12$

The equation above is true for all x, where a and b are constants. What is the value of ab ?

A. 18

B. 20

C. 24

D. 40

2.34 08605-DC · Hard

$(7532+100y^2)+10(10y^2-110)$

The expression above can be written in the form ay^2+b , where a and b are constants. What is the value of $a+b$?

Student-produced response: _____

2.35 f85f6f30-eece-4b2b-bbc1-5c3690c1998f · Hard

Which expression is equivalent to $\frac{42a}{k}+42ak$, where $k>0$?

A. $\frac{84a}{k}$

B. $\frac{84ak^2}{k}$

C. $\frac{42ak+1}{k}$

D. $\frac{42ak^2+1}{k}$

2.36 8c278545-5668-4346-b11a-ea571647594f · Hard

$$\frac{\sqrt[4]{n^{17}p^5}}{5n^2\sqrt[4]{p^8}}$$

For all positive values of n and p, the given expression is equivalent to which of the following?

I. $\frac{n^{\frac{14}{3}}p^{\frac{5}{3}}}{5np}$

II. $\frac{\sqrt[4]{n^{11}p^2}}{5}$

A. I only

B. II only

C. I and II

D. Neither I nor II

2.37 01383d90-cb64-4777-9f89-45b8dfe8c286 · Hard

$$0.36x^2 + 0.63x + 1.17$$

The given expression can be rewritten as $a4x^2 + 7x + 13$, where a is a constant. What is the value of a?

Student-produced response: _____

2.38 b76e28cc-fee7-47dc-ae66-119a52a95d28 · Hard

If $k - x$ is a factor of the expression $-x^2 + \frac{1}{29}nk^2$, where n and k are constants and $k > 0$, what is the value of n?

A. -29

B. $-\frac{1}{29}$

C. $\frac{1}{29}$

D. 29

2.39 7c3cceff-5961-4825-a23b-44ce88420238 · Hard

Which expression is equivalent to $\frac{y+12}{x-8}+\frac{yx-8}{x^2y-8xy}$?

A. $\frac{xy+y+4}{x^3y-16x^2y+64xy}$

B. $\frac{xy+9y+12}{x^2y-8xy+x-8}$

C. $\frac{xy^2+13xy-8y}{x^2y-8xy}$

D. $\frac{xy^2+13xy-8y}{x^3y-16x^2y+64xy}$

2.40 022991-DC · Hard

$$\frac{2}{x-2}+\frac{3}{x+5}=\frac{rx+t}{(x-2)(x+5)}$$

The equation above is true for all $x>2$, where r and t are positive constants. What is the value of rt ?

A. -20

B. 15

C. 20

D. 60

2.41 924baff1-d1d9-4091-910f-6e75684ca3a2 · Hard

The expression $6x^4+31x^2+35$ can be rewritten as $3x^2+a2x^2+b$, where a and b are positive integers, or as $3x^2+c2x^2+d$, where c and d are positive nonintegers. What is the value of $a+c$?

Student-produced response: _____

2.42 16d897b1-c597-4b7a-b4ee-ecba3515085e · Hard

Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$?

A. $\frac{1}{x+14x-5}$

B. $\frac{3}{3x-6}$

C. $-\frac{1}{x+14x-5}$

D. $\frac{9}{x+14x-5}$

2.43 763bb625-3667-4ea6-af9f-195d80193bdb · Hard

$$\sqrt[3]{p^2} = t^{\frac{9}{7}}$$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{3n-1}$, where n is a constant, what is the value of n?

Student-produced response: _____

2.44 00489-DC · Hard

The expression $\frac{1}{3}x^2 - 2$ can be rewritten as $\frac{1}{3}(x-k)(x+k)$, where k is a positive constant.

What is the value of k ?

A. 2

B. 6

C. $\sqrt{2}$

D. $\sqrt{6}$

2.45 00094-DC · Hard

Which of the following is equivalent to $\left(a+\frac{b}{2}\right)^2$?

A. $a^2+\frac{b^2}{2}$

B. $a^2+\frac{b^2}{4}$

C. $a^2+\frac{ab}{2}+\frac{b^2}{2}$

D. $a^2+ab+\frac{b^2}{4}$

2.46 e7cb9147-ae39-47e9-a4cb-da0cfbf0b250 · Hard

The expression $6\sqrt[5]{3^5x^{45}}\cdot\sqrt[8]{2^8x}$ is equivalent to ax^b, where a and b are positive constants and $x > 1$. What is the value of $a + b$?

Student-produced response: _____

2.47 77bede4f-84ec-4063-9b73-a0d1bd0cff6b · Hard

Which of the following expressions is(are) a factor of $3x^2 + 20x - 63$?

I. $x - 9$
II. $3x - 7$

A. I only

B. II only

C. I and II

D. Neither I nor II

2.48 262f0667-b95d-4adf-bdd9-ff689f8a3811 · Hard

If $4^{8c} = \sqrt[3]{4^7}$, what is the value of c?

Student-produced response: _____

2.49 902558e4-018d-442f-a571-ef4e6c7c9fae · Hard

$$9x^4 + 8x^3 - 7x^3 + 5x^2 - 3x^2 + x$$

The given expression can be written in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. What is the value of a?

A. -8

B. -2

C. 2

D. 8

2.50 08243-DC · Hard

In the expression $3(2x^2+px+8)-16x(p+4)$, p is a constant. This expression is equivalent to the expression $6x^2-155x+24$. What is the value of p ?

A. -3

B. 7

C. 13

D. 155

2.51 028934-DC · Hard

$$\frac{x^2-c}{x-b}$$

In the expression above, b and c are positive integers. If the expression is equivalent to $x+b$ and $x \neq b$, which of the following could be the value of c ?

A. 4

B. 6

C. 8

D. 10

2.52 4374fa03-6222-4e03-a842-f2d7a77397bb · Hard

Which expression represents the product of $x^{-6}y^3z^5$ and $x^4z^5 + y^8z^{-7}$?

A. $x^{-2}z^{10} + y^{11}z^{-2}$

B. $x^{-2}z^{10} + x^{-6}z^{-2}$

C. $x^{-2}y^3z^{10} + y^8z^{-7}$

D. $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$

2.53 6291292e-680f-4ffb-9f2e-4bf638eaee8d · Hard

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

A. $3x^2 + 7x + 14b$

B. $3x^2 + 28x + 14b$

C. $3x^2 + 42x + 14b$

D. $3x^2 + 49x + 14b$

2.54 08224-DC · Hard

If a and c are positive numbers, which of the following is equivalent to $\sqrt{(a+c)^3} \cdot \sqrt{a+c}$?

A. $a+c$

B. a^2+c^2

C. $a^2+2ac+c^2$

D. a^2c^2

2.55 baae72cd-ec86-42e7-81a6-5f31c096baea · Hard

The expression $3x - 2319x + 6$ is equivalent to the expression $ax^2 + bx + c$, where a, b, and c are constants. What is the value of b?

Student-produced response: _____

2.56 4feb8c4d-eb86-4f9e-a21a-b22033b4381b · Hard

$$-14(5x - 2)^2 + 6(5x - 3)^2$$

The given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, where a, b, and c are constants. What is the value of $a + b + c$?

A. -612

B. -306

C. -102

D. -17

2.57 08128-DC · Hard

If $a = c + d$, which of the following is equivalent to the expression $x^2 - c^2 - 2cd - d^2$?

A. $(x + a)^2$

B. $(x - a)^2$

C. $(x + a)(x - a)$

D. $x^2 - ax - a^2$

2.58 4b8535d8-7c5d-4bfd-9256-245ea6bb505f · Hard

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as $hx + kx + j$, where h, k, and j are integer constants. Which of the following must be an integer?

A. $\frac{b}{h}$

B. $\frac{b}{k}$

C. $\frac{45}{h}$

D. $\frac{45}{k}$
